Lecture 4: Inference for Subpopulation Proportions and Differences

September 4, 2023

Review

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- We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.
- We also demonstrated that we could predict how close we might by using confidence intervals.
- We explored a bootstrapping approach and a plug-in approach to producing confidence intervals.

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- Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
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- In this lecture, we discuss learning subpopulation proportions and differences between subpopulation proportions based on a random sample (with replacement) from the population.
- Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

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- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout. How might we sample?

2020 Turnout in Georgia i.i.d. sample

- in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.

2020 Turnout in Georgia i.i.d. sample

- in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.
- after the election, we found that the turnout rate among the 30% of registered voters who were black was .74 and among the 70% of non-Black registered voters was .68.

Four Questions

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3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?

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2. Could we have predicted how close we would get to .74 before the election?
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?
4. Could we have predicted how close we would get to a .06 turnout gap before the election?

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\overline{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
	1,0	1,1	...	0,0	.68	.73

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

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1	1,0	1,1	...	0,0	.68	.73
2	0,0	1,0	...	1,1	.70	.72

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Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\overline{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.73
2	0,0	1,0	...	1,1	.70	.72
3	1,0	1,1	...	0,1	.75	.76
4	1,1	1,0	...	0,0	.73	.73
5	0,0	0,0	...	1,1	.69	.71
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1 mil	1,0	0,1	...	1,1	.74	.75
⋮	⋮	⋮	⋮	⋮	⋮	⋮

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

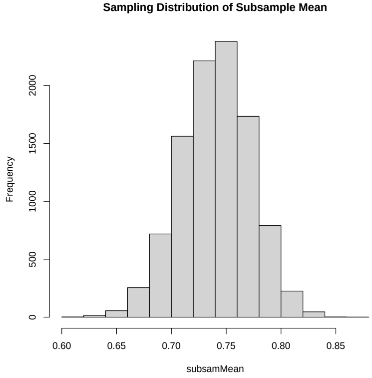
Simulating the Sampling Distribution

ID	1	2	3	4	...	...	...	...	7,233,584	Mean
y_{ID}	0	1	0	1	...	...	...	...	1	.70
x_{ID}	0	0	1	0	...	...	...	...	1	.30

For each row of the sampling distribution,

```
dfpop = data.frame(y,x)
ind = sample(1:7233584, size=625, replace=TRUE)
dfind = dfpop[ind,]
subsamMean = mean(dfind$y[dfind$x == 1])
# alternative approach
#subsamMean = sum(dfind$y*dfid$x)/sum(dfind$x)
```

```
S = 10000
subsamMean = rep(NA,S)
for( s in 1:S ){
ind = sample(1:N,size=625,replace=T)
dfind = dfpop[ind,]
subsamMean[s] = mean(dfind$y[dfind$x==1])
}
```



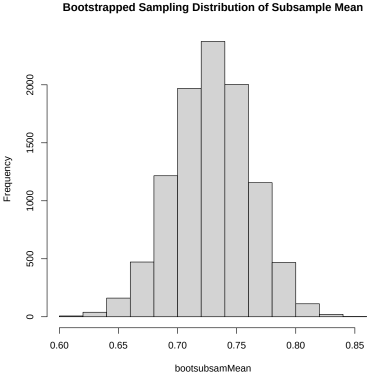
Simulating the Bootstrapped Sampling Distribution

i	1	2	...	625	$\bar{y}_{625}$
ID	5093883	5473235	...	3042073	
y_i	1	1	...	0	.68
x_i	1	0	...	0	.29

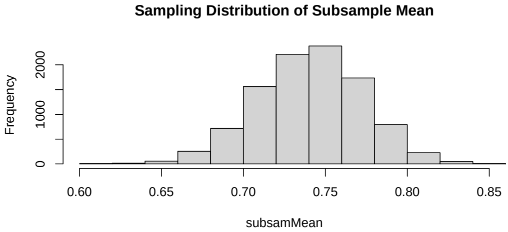
Starting with the sample (first row of sampling distribtution), each row of the bootstrapped sampling distribution is generated as the following:

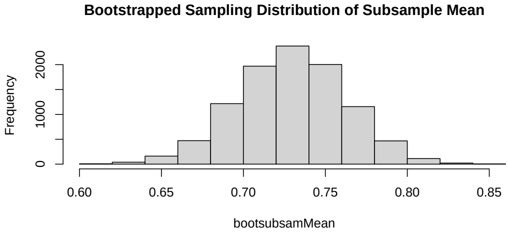
```
bootind = sample(1:625, size=625, replace=TRUE)
dfboot = dfsam[bootind,]
subsamMean = mean(dfboot$y[dfboot$x == 1])
# alternative approach
#subsamMean = sum(dfboot$y*dfboot$x)/sum(dfboot$x)
```

```
B = 10000
bootsubsamMean = rep(NA,B)
for( b in 1:B ){
bootind = sample(1:625,size=625,replace=T)
dfboot = dfsam[bootind,]
bootsubsamMean[b] = mean(dfboot$y[dfboot$x==1])
}
```



Sampling Distribution vs Bootstrapped Distribution






```
> quantile(bootsubsamMean,prob =c(.025,.975))
     2.5%     97.5%
0.6632653 0.7926845
```

Four Questions

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1. .73 is close to the Black turnout rate of .74. Did we get lucky? Maybe a little, sampling distribution from .65 to .85
2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?
4. Could we have predicted how close we would get to a .06 turnout gap before the election?

Black Turnout Minus non-Black Turnout

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i} - \frac{\sum_{i=1}^{625} Y_i (1-X_i)}{\sum_{i=1}^{625} (1-X_i)}$
	1,0	1,1	...	0,0	.73	.07

- We now need the sampling distribution of the difference in subsample means.
- Again, we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population, and calculate the difference in proportions.

Black Turnout Minus non-Black Turnout

i	1	2	$\dots$	625		
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1	1,0	1,1	$\dots$	0,0	.73	.07
2	0,0	1,0	$\dots$	1,1	.70	.03

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1	1,0	1,1	$\ldots$	0,0	.73	.07
2	0,0	1,0	$\ldots$	1,1	.70	.03
3	1,0	1,1	$\ldots$	0,1	.76	.09
4	1,1	1,0	$\ldots$	0,0	.73	.06
5	0,0	0,0	$\ldots$	1,1	.71	.04
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	1,0	0,1	$\ldots$	1,1	.74	.07
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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For each row of the sampling distribution of differences,

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dfpop = data.frame(y,x)
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dfind = dfpop[ind,]
diffMean = mean(dfind$y[dfind$x == 1]) -
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Write down the code for the ????.

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dfind = dfpop[ind,]
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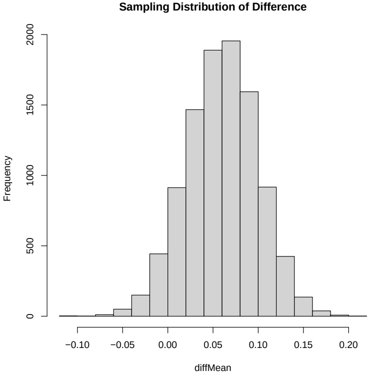
mean(dfind$y[dfind$x==0])
}
```



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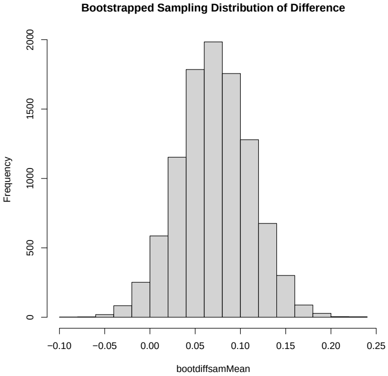

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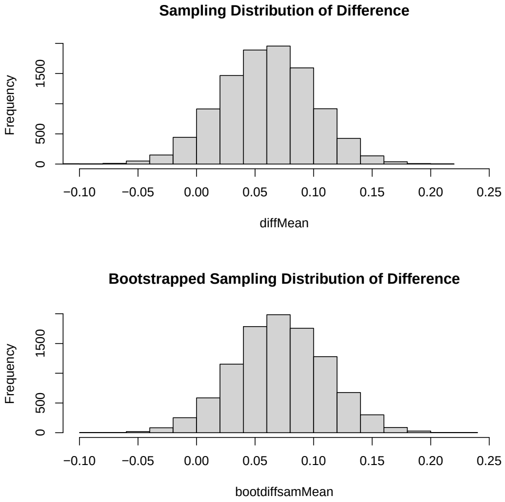


```
> quantile(bootdiffsamMean,prob =c(.025,.975))
        2.5%        97.5%
-0.006698302  0.148697667
```

Four Questions

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2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky? A bit, sampling distribution can go from 0 to .15
4. Could we have predicted how close we would get to a .06 turnout gap before the election? Yes, CI gives correct sense of uncertainty.

Connection to Linear Regression Slopes

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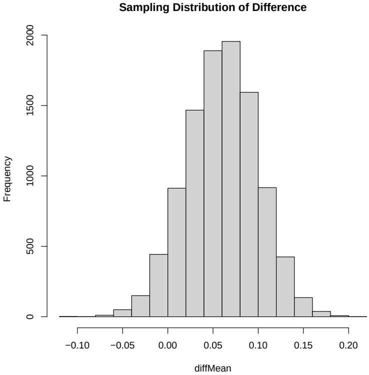
- We can use this concept to re-write the sampling distribution of difference and bootstrap code for differences.
- On the next two slides, see if you can fill in the ? marks.

```
S = 10000
diffMean = rep(NA,S)
for( s in 1:S ){
ind = sample(1:N,size=625,replace=T)
dfind = dfpop[ind,]
diffMean[s] = lm(? ~ ?, data=?)$coef[?]
}
```



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bootdiffsamMean[b] =  ??????
}
```

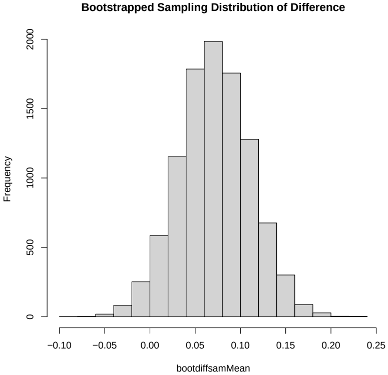

Outputs from the lm() function

```
summary(lm(y~x,data=dfsam))

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.65766    0.02215   29.69   <2e-16 ***
x            0.07162    0.04116    1.74   0.0823 .
---
```

From the *x* row:

- Estimate: 0.07
- Std. Error: 0.04
- t value: 1.74
- p value: 0.08
- stars and dots

Estimate, Std. Error, and t value

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sd(bootdiffsamMean)
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The t values is just the ratio of the estimate and the estimated Std. Error

$$\frac{.07}{.04} = 1.75$$

Stars and Dots

Stars and Dots indicate different levels of significance:

- If the 90% CI doesn't cover 0, but the 95% does, it gets a dot
- If the 95% CI doesn't cover 0, but the 99% does, it gets a star
- If the 99% CI doesn't cover 0, but the 999% does, it gets two stars
- If the 999% CI doesn't cover 0, it gets three stars

```
quantile(bootdiffsamMean,prob =c(.05,.95))
         5%          95%
0.006350511 0.137272113
quantile(bootdiffsamMean,prob =c(.025,.975))
        2.5%        97.5%
-0.006698302  0.148697667
quantile(bootdiffsamMean,prob =c(.005,.995))
       0.5%       99.5%
-0.02982954  0.17289784
quantile(bootdiffsamMean,prob =c(.0005,.9995))
      0.05%      99.95%
-0.05634436  0.20425806
```

P value

The p-value is 1 - confidence level for which the edge of the interval is on (or closest to) zero

```
quantile(bootdiffsamMean,prob =c(.035,.965))
        3.5%         96.5%
-0.000494944  0.143326111
quantile(bootdiffsamMean,prob =c(.036,.964))
        3.6%         96.4%
0.0002851073 0.1428813860
```

Somewhere close to 0.072. Again a little different than from the table due to boostrapping producing a better p-value.

Summary

...